

KUSHAGRA SINGHAL

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SUMMARY

Founding Engineer and Scrum Master with over 1 year of experience building 5 AI-powered products across Education, Advertising, Fleet Intelligence, Defense, and Conversational AI. Proficient in **Python, FastAPI, React, Next.js, LangChain, YOLOv10, Docker, and PostgreSQL** with hands-on experience in **multi-agent AI systems, RAG pipelines, computer vision, microservices, CI/CD, and QA automation**. Seeking Backend, Full-Stack, or AI Engineer roles.

WORK EXPERIENCE

Founding Engineer - Datavedam, Bangalore | Dec 2025 - Present

- Spearhead **5 AI-driven products** at a pre-revenue startup, owning full-stack delivery, DevOps, QA strategy, and Scrum ceremonies.
- ShikshaVedam**: Developing a school ERP with a multi-agent **Virtual Principal** (FastAPI, LangChain, RAG, OCR, Keycloak RBAC); serving 500+ students at Ashoka World School.
- PracharVedam**: Designed a **6-microservice go-to-market platform** for AI ad generation, brand analysis, and lead scoring.
- PravahVedam**: Designing a fleet intelligence system with **11 AI agents**, vehicle-to-vehicle communication, drowsiness detection, and 30+ dashboards using Next.js and React Native.
- AstraVedam**: Pioneering a counter-drone defense solution with **YOLOv10** object detection (95% mAP), computer vision geofencing, instant alerts, and custom RF jamming hardware.
- Vebot**: Launched an AI chatbot service deploying virtual assistants on **WhatsApp, Telegram, and Discord** with encryption and 60-second onboarding.
- Established **QA processes** across 5 products with **Robot Framework** and pytest; executed test suites via GitHub Actions and tracked defects in Jira.

Data Analyst Intern - Nineleaps, Bangalore | Mar 2025 - Aug 2025

- Delivered **4 analytics modules** and BI dashboards for **30+** stakeholders; optimized SQL data pipelines processing **1M+ rows**, improving performance by **35%**.
- Configured 3 CI/CD pipelines with automated testing (pytest) and introduced an **LLM agent** (LangGraph) for Git issue triage and metric suggestions.
- Authored standard procedures and trained **5 teammates** on SQL optimization, DAX, and QA workflows.

PROJECTS

Lakshya Vedh (Sports AI Coach) - Datavedam (Bihar Sports Program)

- Created an automated **biomechanical analysis** engine (**MediaPipe**) for pose estimation and **center-of-gravity scoring** across 50+ athletes; presented to the Bihar Health Minister.
- Integrated **Ollama (Llama 3.1 8B)** for bilingual coaching via Gradio; tailored for offline deployment in rural health camps and schools.

BroAI - Converged Cognitive Architecture - Datavedam

- Devised a production-ready **AI agent memory module** (PostgreSQL, vector search, **ACID guarantees**); deployed as a grievance reporting portal processing 1,000+ requests.
- Implemented **hybrid search** with 92% retrieval accuracy, reciprocal rank fusion, memory decay, PII masking, and a **transactional outbox pattern**.

AI Issue Triage and Code Remediation Assistant - Personal Project

- Constructed a **LangChain and LangGraph** agent analyzing Git issues and code changes to propose fixes and unit tests across 3+ repositories.

MAGI Generative AI - Nineleaps (Uber client)

- Benchmarked **4+ LLMs** for evaluation tasks; improved model reliability and consistency by **15%**.

Anarres LLM Analytics - Nineleaps (Uber client)

- Streamlined SQL querying via an AI pipeline (BigQuery, Google Sheets) achieving **95% precision**; reduced manual effort by **40%**.

AI Email Response Agent - Hackathon

- Prototyped a React, FastAPI, and LangGraph assistant in 48 hours with reusable UI components and unit-tested backend services.

TECHNICAL SKILLS

- Languages**: Python, C#, SQL
- Software Engineering**: Object-oriented programming, SOLID principles, SDLC, microservices, design patterns, code reviews
- Testing and QA**: Manual, unit (pytest, JUnit), regression, integration, smoke testing, Robot Framework, test automation, defect tracking
- AI and ML**: LangChain, LangGraph, RAG, YOLOv10, OpenCV, MediaPipe, Ollama, Gemini, Vertex AI, prompt engineering
- Backend**: FastAPI, REST APIs, WebSockets, Gradio, Alembic (DB migrations), Poetry, Razorpay
- Frontend**: React, Next.js, Vite, Tailwind CSS, UI component libraries, React Native, Framer Motion
- Databases**: PostgreSQL, Redis, ClickHouse, BigQuery, SQLAlchemy, vector databases
- DevOps**: Docker, Docker Compose, Git, GitHub Actions (CI/CD), Keycloak, Nginx
- BI and Data**: Power BI, Tableau, ETL and ELT pipelines, KPI design, A/B testing
- Agile and Scrum**: Scrum Master, cross-functional collaboration, team mentoring, Jira, Confluence, sprint planning, backlog grooming

EDUCATION

Christ University, Bangalore | Aug 2021 - May 2025

Bachelor of Technology in Computer Science and Engineering

CERTIFICATIONS

- Google Data Analytics Professional Certificate** - Coursera
- AI Agents with LangChain and LangGraph** - IBM